

Combined MPLPB

A Provenance Ledger with a Refusal: One Format, One Rule, and What the Kill Test Measured

Field	Value
Document ID	MPLPB-COMBINED-017 (proposed; the next free number after MPLPB-IMAGE-016 in the supplied material, not yet allocated by the owner's index)
Category	Specification / Reference Implementation
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Owner	Mitchell D. McPhetridge, Independent Researcher
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Scope	The single record format and the single answer rule that the MPLPB papers and implementations share; a standard-library reference implementation of exactly that rule; how each earlier part maps onto it; and one run of the kill test with its limits
When to use	Starting a new corpus and wanting one format instead of five; reading a corpus written by any earlier MPLPB tool; deciding what the formula does and does not guarantee; checking what the kill test showed
Supersedes	MPLPB-COMBINED-017 v1 (2026-10-07T00:00Z), retained under docs/superseded/. No earlier paper or implementation is superseded; each stays current within its own scope and is carried under parts/

Related. MPLPB as a Local Web (MPLPB-LOCAL-008 v4). Separate, Allowed to Chat, and Overridden by Scope (MPLPB-SEP-007). Smart Local MPLPB (MPLPB-SMART-011 v1). MPLPB and the Cost of Visibility (MPLPB-COST-012 v2). MPLPB at Swarm and Enterprise Scale (MPLPB-SWARM-013 v3). Networked MPLPB. MPLPB Image (MPLPB-IMAGE-016 v2). Multi-Platform Linked Public Building (v3) and the other chapters of *Continuity Without Memory*.

Implementation. This repository. Package `mplpb_combined`, 97 tests, standard library only, no network, no model, no database.

What changed in v2. The first run of the kill test exposed errors in the reader. They were fixed, the test was rerun, and a second probe set was written to check the fix. `docs/FIXLOG.md` records each error, its cause and its fix. Sections 3, 6 and 8 are rewritten; the first run's results are kept under `killtest/v1/`.

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Abstract

The MPLPB papers describe one idea several times at several scales: a body of knowledge kept as complete pages that declare what they are, read by something that is not allowed to answer from anywhere else. By the time the idea had a local form, a reader, a cost argument, a swarm form, a networked form and an image form, it also had five implementations, three spellings of the same metadata, and two different procedures for retiring a page.

This paper states the part they have in common and nothing else. It is a closed local ledger of pages that answers only from those pages. Each page carries its own id, scope, status, hash, parent and depth. The reader crawls the folder. If one current page owns the question, it returns that page. If two own it, it stops. If none own it, it says not in the corpus. A change writes a new page and retires the old one. A derivation writes a new page and adds one to the depth.

The reference implementation is about twelve hundred lines of standard-library Python for the format, the ledger and the reader, plus a command line and the test harness. It reads the example sites of four earlier implementations after adding hashes to them, and it carries all five implementations unchanged in behaviour under parts/.

The formula proposed a proof: strip the scope fields, run the same questions, and show the refusal disappears. That test has now been run, the reader has been corrected against it, and it has been run again.

The first run found a bug. The reader refused every question phrased from a page's prose, and one stray word declared on another page was enough to stop it. Both are fixed: ownership is now decided by declared scope first, as a majority of the question's words, and by the page's own text only when scope owns nothing. On the 56 questions the fix was developed against, the reader went from 41 right to 50, with no wrong page returned in either run.

A second set of 30 questions, written after the fix and run once, is the honest measure. There the fixed reader is right 20 times, against 12 for the first reader, and it returns a wrong page 3 times, against 4. Every one of those wrong returns is a question about something the corpus covers, asked from an angle it does not. So the first run's clean record was a property of its questions. The declared fields make the reader far better than search over the same pages with the fields stripped, which is wrong 20 times in 30. They do not make it safe.

1. The Formula

A record is

```
R = (id, scope, status, hash, derived_from, origin_depth)
```

Status is current or retired. A question q is answered only from current records whose scope owns q:

A(q) = R	if exactly one current R owns q
ambiguous	if two or more own q
not in corpus	if none own q

No fill from outside the folder. No average of two owners.

Derivation does not copy the hash. It writes a new record and increments depth:

```
R[n+1].derived_from = R[n].id
R[n+1].origin_depth = R[n].origin_depth + 1
```

A human source starts at depth 0. A revision retires R[n]; it does not overwrite it. A profile may refuse any record with depth above a set limit. A hub may point at the owner; it may not answer as the owner.

Existing parts, one composition rule: do not blend, do not erase, count the generations.

Everything below is what had to be decided to make those sentences executable, and each decision is marked as one.

2. The Format

A record is one HTML file. Its <head> carries <meta name="mplpb:..."> tags and its <body> carries the words. Opened in a browser it is a page. Nothing else is needed to read it.

Field	Required	Meaning
document-id	yes	Allocated, never chosen, never reused
scope	yes	One sentence saying what the page covers
status	yes	current or retired
hash	yes	SHA-256 over the title, the body, and every field except status and hash
derived-from	when derived	Parents, each as ID@sha256:...
origin-depth	yes	Generations between this page and a source
when-to-use	no	The situations in which someone should land here, in the asker's words
supersedes	when revising	The page this one replaces, as ID@sha256:...
origin	no	human or machine; absent means human
ratified-by	when ratified	The name of the person who stands behind a derived page
kind, points-to	hubs only	pointer, and the folder it points at
owner, updated, category	no	Carried, hashed, not interpreted

Three decisions in that table are new and deliberate.

The hash excludes status. Retiring a page changes one line in its head. Because that line is outside the hash, a retired page still verifies, and "it does not overwrite it" becomes checkable: the hash of the retired page is the hash it had when it was current.

References are pinned. A child names its parent's id and its parent's hash. Editing a parent, or swapping in a different page under the same id, breaks every descendant's pin. This is integrity without keys. It shows a page was not changed after it was written. It does not show who wrote it; signatures are the Networked part's job and stay there.

One spelling. The earlier tools wrote `mplpb:document-id` and `mplpb:id`, `origin-depth` and `origin_depth`, `updated` and `version`. All are read as the same field. New pages are written with hyphens.

A file under the root with no `document-id` is not a record and is never read. An index page is recognised by name, category or kind and never owns a question. `index.html` is rebuilt from the records for people to browse and carries no authority.

3. What "Owns" Means

The formula leaves one word undefined, and it is the word everything depends on. The earlier reader, Smart Local, defines ownership as a ranked full-text score in which one spoke leads the runner-up by 1.5 times, and calls that number the most arbitrary in the package. This implementation defines ownership without a score, in two steps. Scope always goes first.

- 1 **Scope.** A page owns the question when its `scope` and `when-to-use` fields declare more than half of the question's content words.
- 2 **Prose.** Only if no page owns the question by scope: a page owns it when every content word appears somewhere on the page.

At either step, when two pages own the question and one's matched words strictly contain the other's, the more specific page stays and the other drops out. Then the rule is the formula as written: one owner returns, two or more is ambiguous, none is not in the corpus.

Content words are lower-cased, stripped of a short stop list, and stripped of plural, *-ing* and *-ed* endings, so that `firing`, `fired` and `fire` are one word. There is no synonym table.

Three things about this definition are worth stating.

Scope overrides prose. If any page owns the question by what it declares, no page's prose is consulted. This is MPLPB-SEP-007 applied inside one corpus: declared scope decides, and prose only speaks when scope is silent. It is also "narrow before broad" from MPLPB-SMART-011 section 4, with the two passes in the opposite strictness: the declared pass needs a majority because declarations are short and deliberate, and the prose pass needs every word because prose is long and accidental.

A prose match says it is one. The citation of a page returned at step two ends matched on `prose`, not on `declared scope`. A reader of the answer can see that the author did not predict the question.

One constant remains: more than half. Version 1 had a second, a per-profile coverage share, and a rule that ignored words most pages declare. The first run showed the coverage dial moving two answers across its whole range, and both are gone.

The first version of this section defined ownership as declaring *every* word that any page declared, and matched no prose at all. Both choices were wrong in ways the first run made plain, and `docs/FIXLOG.md` entries E1 and E2 record them.

The decision function takes the question's words and two maps of sets, and is shared by the reader and by the kill test's control arm, so the two cannot drift apart.

4. Writing

Four verbs write to a ledger. None of them changes a word of an existing page.

write allocates the next id for a prefix, renders the page with its hash, and puts it in the folder. It refuses if the file exists.

revise writes a new page that names the old one in `supersedes`, then sets the old page's status line to `retired`. The order is the point. The reader treats any page named in a `supersedes` field as retired whatever its own status line says, so a crash between the two steps leaves a ledger that still reads correctly, and the validator reports the unfinished step. Smart Local's notebook retires first and writes second; a crash there leaves no current page, and the reader says "not in this corpus" about something that exists.

Retired pages are not moved. The earlier tools move them to `_log/superseded/` and treat a retired page elsewhere as a fault. Here location carries no meaning and status does, which removes a step that can fail and a class of broken relative links.

derive writes a new page naming its parents, pinned, and sets depth to one more than the deepest of them.

ratify is a revision that changes no words. It writes a new page with the same body and lineage, the name of the person standing behind it, and depth 0. The origin stays `machine`. The citation then reads `origin machine d0 · ratified by Name`, which says more than rewriting the origin to human would.

Writers take a lock file for the duration of one write. A lock older than thirty seconds belonged to a writer that died. This is enough for a person and a few scripts. It is not the lease-and-fencing discipline of MPLPB-SWARM-013 section 4, which remains the answer when writers are many.

Every write and every retirement appends a line to `_log/ledger.jsonl`. Each line carries the hash of the line before it, so the log cannot be edited in the middle without the chain breaking.

5. Depth

The formula says a derivation adds one. Five cases needed settling.

Case	Depth
Human page with no parents	0
Machine page with no parents	1. The loop cannot award itself a human signature (MPLPB-SWARM-013 section 5.3)
Derived page, whoever derives it	1 + the deepest parent
Revision	At least the depth of the page it replaces
Ratified page	0, with the ratifier's name on the page

The fourth row closes a path the earlier tools leave open. If a revision took its depth only from its author, a person tidying a depth-3 page would produce a depth-0 page without ever saying they stood behind it. Here a reset happens only through `ratified-by`, which is a name on the page and a line in the log.

The reader does not trust the declared number. It recomputes depth from the lineage and serves the larger of the two. A page that understates its depth is an error to the validator and is not believed by the reader.

What this does not do is unchanged from the swarm paper: it does not detect an invented claim. A writer that marks a machine page human, or types a name into `ratified-by` without reading, defeats it. Depth is a count of declared generations.

6. Profiles and Hubs

A profile is two settings: the deepest record it will serve, and whether the prose step may run.

Profile	Max depth	Prose step
lab	2	on
internal (default)	1	on
external	0	off

A page over the limit is removed before ownership is decided. When it would have owned the question, the refusal says so and gives the page's id and depth. `external` decides by declared scope or not at all, because a customer should not be handed a page on the strength of words its author never offered as its subject. `answer_without_retrieval` has no setting here because there is no code path that produces words other than a page's own.

A hub is a ledger whose records are pointers. A pointer declares a scope like any page and names another folder. When a pointer owns a question, the reader asks the same question in that folder and returns what that folder returns, with the route attached. When no pointer declares the question, each folder the hub points at is asked in turn; exactly one may answer, and two answering is ambiguity between corpora, which the hub names and stops on. A folder that cannot be reached produces a refusal that names it. No path returns a pointer's own body, and no path merges two folders' answers. This is MPLPB-SEP-007 one level up, as the swarm paper specified, without the network that Networked MPLPB adds.

7. How the Parts Map

Nothing is superseded. Each part keeps what only it does.

Part	What the combined core takes from it	What stays in the part
Public MPLPB (v3) and the complete-artifact header	The fields: identity, scope, trigger conditions, status, supersession	The public web as substrate; reconstruction by a search-capable assistant; the recoverability, stability and validation questions
Rabbit Hole Internet Indexer	Nothing executable. The core has no index	An indexing and retrieval layer in front of a corpus, with embeddings as a replaceable component

Part	What the combined core takes from it	What stays in the part
Local Web (LOCAL-008) and the Local Mirror	Pages as plain HTML under one root; bounded reading; metadata in the head	Link-graph crawl, typed links, the eight structural checks, the boot block
Separation and Precedence (SEP-007)	The rule itself: scope decides, ties stop, nothing merges	The argument for it
Mode controller (MODE-008)	Nothing. The core has one behaviour	Modes, overrides, the hard-reality priority
Smart Local (SMART-011)	Refusal as the default; provenance as part of the hit; revision as supersession; narrow before broad	FTS5 ranking, the 1.5× margin, the console, the browser view, teaching
Cost of Visibility (COST-012)	The constraint: consumer hardware, no service	The cost model and the comparability claim
Swarm (SWARM-013)	Origin depth, ratification, profiles, allocated ids, routing between corpora without merging	Leases with fencing tokens, the registry, health metrics, the graded ablation
Networked MPLPB	Hubs that point and do not answer; unavailable is never impersonated	Ed25519 signatures, nodes, mirrors, transport
MPLPB Image (IMAGE-016)	The principle that the page is the control object and the file is the artifact	Pixel evidence, attestation, the learner
Public Experiment (TEST-001, TEST-002)	Pass criteria fixed before a run	The replication protocol and trial ledger

On search and the public web. The formula is a property of a closed reader. A folder read by this package obeys it. The same pages published to the open web are read by somebody else's search engine and somebody else's model, and nothing stops either from blending two pages or filling a gap. Public MPLPB can make the right answer findable and can make the fields travel with it. It cannot enforce the refusal. That limit follows from the formula's first word, and the public papers already separate recoverability from validation for the same reason.

A ranker, lexical or semantic, can still sit in front of this core to propose candidates on a large corpus. It may narrow which pages are considered. It may not make the decision.

8. The Kill Test

8.1 What was run

One corpus, `examples/studio`: 25 pages about running a pottery studio. 21 are current and 4 are retired. Three of the current pages are machine-written: one at depth 1, one at depth 2, and one ratified back to depth 0.

Three arms read the same pages.

- **scoped** is the reader in this package at the `internal` profile.

- **stripped** is the same rule and the same code after every `mplpb:` field is removed. With nothing declared, the scope step finds nothing and only the prose step can decide; with no status left, retired pages cannot be told from current ones.
- **top1** is what a plain search does with the stripped pages: return the page sharing the most words with the question, and refuse only when nothing shares any. It has no way to say ambiguous.

Scoring is mechanical. An answer is correct when the outcome matches and, for a return, the page matches.

There are two probe sets, and they are not the same kind of evidence.

- The **first set** is 56 questions, frozen by hash before the first run. The reader was then corrected against its failures. It is a development set now. Its hash is unchanged, and its numbers show that the bugs are fixed and nothing more.
- The **held-out set** is 30 questions, written after the reader was corrected and before the corrected reader was run on any of them, frozen by hash, and run once. Nothing was changed afterwards.

8.2 The first set: before and after the fix

Reader	Correct	Wrong page returned	Wrong refusal
scoped, version 1	41	0	15
scoped, version 2	50	0	6
scoped, version 2, prose step off	43	0	13
stripped	26	4	26
top1	16	40	0

By class, for version 2:

Class	n	scoped	stripped	top1	Note
Asked in a page's declared words	10	100.0%	10.0%	10.0%	
Asked from a page's prose	8	87.5%	87.5%	100.0%	was 0.0% in version 1
Declared situation plus unfamiliar words	8	62.5%	0.0%	12.5%	
Two current pages both answer	6	100.0%	16.7%	0.0%	circular for top1
The answer was revised	6	100.0%	0.0%	16.7%	circular for both stripped arms
Only owner is over the depth limit	2	100.0%	50.0%	0.0%	circular for both stripped arms
Nothing to do with the corpus	6	100.0%	100.0%	83.3%	control at ceiling
Studio question no page covers	10	80.0%	100.0%	0.0%	control at ceiling
All 56	56	89.3%	46.4%	28.6%	
Non-circular classes only	42	85.7%	57.1%	35.7%	

Of the 28 pages the fixed reader returned, 21 were decided by scope and 7 by prose. Six probes still fail, and `docs/FIXLOG.md` lists each with its reason. Each could be made to pass by moving a threshold or adding a stop word. Section 8.3 is why that was not done.

8.3 The held-out set

Reader	Correct	Wrong page returned	Wrong refusal
scoped, version 1	12	4	14
scoped, version 2	20	3	7
scoped, version 2, prose step off	16	3	11
stripped	10	1	19
top1	10	20	0

Class	n	scoped	stripped	top1	Note
Asked in a page's declared words	8	87.5%	0.0%	25.0%	
Asked from a page's prose	8	37.5%	25.0%	75.0%	
Two current pages both answer	3	100.0%	0.0%	0.0%	circular for top1
The answer was revised	3	66.7%	0.0%	0.0%	circular for both stripped arms
Studio question no page covers	8	62.5%	100.0%	25.0%	control at ceiling
All 30	30	66.7%	33.3%	33.3%	
Non-circular classes only	24	62.5%	41.7%	41.7%	

The three wrong returns:

Question	Returned
who is allowed to use the gas kiln	the gas kiln firing schedule
how do I center clay on the wheel	the wedging page
how do I pull a handle	the drying page

8.4 Reading it

The fix is real. On questions it was not developed against, the corrected reader is right 20 times where the first was right 12, and it makes fewer wrong returns, not more. The prose step accounts for 4 of the 20 and adds no wrong return: with it switched off the count of wrong returns is still 3.

The first run's clean record was luck. Version 1 returned no wrong page on the first 56 questions and four on the next 30. Version 1 of this paper said the declared fields "bought exactly that trade". They did not buy it outright. The first probe set did not contain the kind of question that breaks it.

The kind of question that breaks it is the adjacent one. All three wrong returns ask about something the corpus covers, from an angle it does not: who may use the gas kiln, how to center clay, how to pull a handle. The page that declares the subject owns the question, by scope, exactly as designed. No rule that matches words can tell a firing schedule for the gas kiln from a rota for it. This is recorded as FM-C8, and it is the main thing this version adds to what is known.

Declared scope still does most of the work. Stripped of its fields, the same code is right 10 times in 30 and refuses 19 it should not. Plain search over the stripped pages returns a wrong page 20 times in 30. The reader with declared scope is wrong in that way 3 times. That gap survived a set of questions the reader

was not tuned on.

The refusal still does not "disappear" when the fields are stripped. It was true in the first run and it is true in both sets now. With the same rule, stripping the fields makes the reader refuse far more, not less. The refusal disappears only against a reader that never had one.

Prose questions are half fixed. 7 of 8 on the first set and 3 of 8 on the held-out set. The prose step needs every word of the question on one page, and people do not ask in the page's exact words. Loosening it lets wrong pages through, which is the top1 arm.

8.5 Why the fixing stopped

The instruction was to fix, rerun, and repeat until the results match. The bugs are fixed and the reruns are recorded. The results do not match on every probe, and they were not forced to.

The remaining failures on the first set could each be removed by a small adjustment, and the adjustments pull against each other: "at least half" instead of "more than half" passes two more probes and turns a correct refusal into a misplaced one. A reader adjusted until one probe set reads 100% is fitted to that set. The held-out set already shows what such a number is worth: version 1 scored a perfect record on wrong returns over 56 questions and it did not hold for 30 more.

8.6 What this evidence is not

- **It is one author's run.** The corpus, the declared words, both probe sets and the reader were written by one hand, by a language model working from the owner's formula. The held-out set was written blind to the fixed reader's output, not blind to the corpus or the rule. This is the weakest class of evidence in MPLPB-COST-012 section 4.1.
- **The vocabulary confound of MPLPB-SWARM-013 section 13.6 applies in full.** Declared-word questions succeed because the same author wrote the declarations and the questions.
- **Three classes are circular** for at least one arm. They are in the tables for completeness and are excluded from the last row of each.
- **It is small.** 86 questions over 25 pages.
- **It is retrieval, not reconstruction.** The ablation MPLPB-LOCAL-008 section 14 named is still open.

9. The Parts, as Run

Every vendored part was run on Linux with Python 3.12 from this repository's `parts/` directory. `tools/check_parts.py` repeats it.

Part	Its paper says	Ran here
Smart Local	52 tests	52 pass
Swarm	144 tests	145 pass
Networked	61 tests	61 pass
Image v2	not stated	93 pass
Local Mirror	no test suite	its own validator exits clean

Part	Its paper says	Ran here
Combined (this package)		97 pass

Three things were changed in the vendored copies, and nothing else.

- 1 In Smart Local and the Local Mirror, the directories Site, Tests, Tools and Docs were renamed to lower case, which is what their READMEs and imports already say. As published, Smart Local's tests do not start on a case-sensitive filesystem.
- 2 The Swarm copy is taken from the archive bundled in its repository, not from the repository tree. The tree is missing ablation/v2/arm-structured/deploy/_index.html and carries a stray registry folder inside site/corpus-b; three tests fail there and pass from the archive. The swarm paper says 144 tests; the suite has 145.
- 3 Nested archives and caches were removed.

The example sites of the Local Mirror, Smart Local, Swarm (both corpora) and Networked were each copied, given hashes with seal, and validated in the combined format with no errors. That is the evidence that the format in section 2 is a superset of what the family already writes.

10. Failure Modes

The FM-L and FM-S series stand. These are the ones this consolidation adds or sharpens.

FM-C1 — Unpredicted vocabulary. A page exists, answers the question, and is not returned, because the asker used neither its declared words nor enough of its prose. In version 1 this was 8 of 8 prose questions. The prose step reduced it to 1 of 8 on the first set and 5 of 8 on the held-out set. Detection: log refused questions and read them.

FM-C2 — Incidental claim. An ordinary word in a when-to-use phrase gives its page a claim on questions it has nothing to do with. In version 1 one such word was enough to stop the reader. Under the majority rule it takes more than half the question. Mitigation: write trigger phrases with as few ordinary words as they can carry.

FM-C3 — Status drift. A page named in supersedes still says current. The reader is unaffected. Detection: check C9. It marks a revision that did not finish.

FM-C4 — Derivative crowding. A derived or ratified page redeclares its parent's scope and the parent stops being the sole owner. Detection: ambiguity between a page and its own descendant. Mitigation: a derived page declares what it adds, not what it summarises.

FM-C5 — Self-consistent forgery. Someone edits a page and recomputes its hash. The page verifies. Its children's pins do not, so the forgery is caught only if the page has descendants or the log is intact. A leaf page edited together with the log is not caught by anything in this package. Signatures are the remedy and live in Networked MPLPB.

FM-C6 — Declared origin. A machine page marked human, or a name typed into ratified-by unread. Depth then counts nothing. Not detectable from the files, as the swarm paper already says of FM-S9 and

FM-S12.

FM-C8 — Adjacent question. The corpus covers a subject, the question asks about that subject from an angle no page covers, and the page that declares the subject is returned. Three of 30 held-out questions. It is a wrong return, the failure the design exists to prevent, and it is decided by scope, not prose. Not detectable by any rule over words. Mitigation is authoring: a page can only be kept from owning what it does not cover by saying what it does not cover, and the format has no field for that yet.

FM-C7 — Hub blind spot, traded for a fan-out. A hub whose pointers declare too little would refuse questions its corpora own. The fan-out in section 6 removes that at the price of asking every corpus, which is linear in the number of pointers and makes cross-corpus ambiguity more frequent.

11. Falsifiers

11.1 — No blend (run; passes). No code path returns text that is not one page's own. Tests assert it for the three cases, the split case, hubs and fan-out.

11.2 — No erase (run; passes). After a revision the old page is present, retired, and verifies against the hash it had when current. A crash between write and retire still reads correctly.

11.3 — Count the generations (run; passes). Derivation adds one whoever derives; a revision cannot lower depth; an understated depth is reported and not believed; no question reaches a page over a profile's limit.

11.4 — The kill test (run; reader corrected; rerun; mixed). Section 8. The claim "stripping scope removes the refusal" is contradicted in every run. The claim "declared scope prevents wrong returns" is weakened: 3 wrong returns in 30 held-out questions, against 20 for plain search. It needs repeating with probes written by someone who has not read the corpus, which is the first item of MPLPB-SWARM-013 section 13.5 and still needs one other person and an afternoon.

11.5 — Real vocabulary and adjacent questions (open). Take refused and returned questions from real use of a real corpus. Count how many refusals had an answer on a page (FM-C1) and how many returns were the subject's page for a question it does not answer (FM-C8). The second number decides whether this reader can face anyone who cannot tell.

11.6 — Scale (open). The reader parses every page on every question. That is immediate at 25 pages and has not been measured at 25,000.

11.7 — Old hardware (open). The claim that a 2010 machine can run this is an inference from what the code uses: Python 3.8 syntax, the standard library, no index held in memory. It was run only on current hardware under Python 3.12.

12. What This Does and Does Not Establish

It does not establish:

- that a returned page is true;
- that declared scope beats flat storage in general. On the 24 non-circular held-out probes the margin was 62.5% against 41.7%, on one corpus, by one author;
- that the reader returns no wrong page. It returned 3 in 30 held-out questions;
- that people ask in the words pages declare;
- that the format is tamper-proof. It is tamper-evident for pages with descendants or an intact log, and nothing more;
- that the formula holds on the public web, where the reader is not yours;
- that this package replaces any of the parts.

It does establish:

- that the formula can be implemented exactly, with ownership defined on sets and no score, in about twelve hundred lines with no dependency;
- that one format reads what four earlier implementations write;
- that revision can be made safe against interruption by ordering two steps and by letting the reader trust supersedes over status;
- that depth can be recomputed from lineage, so that the number in a citation is derived and not merely declared, up to the honesty of origin;
- that the errors the first run exposed were bugs and are fixed: on held-out questions the corrected reader is right 20 times in 30 where the first was right 12, with fewer wrong returns;
- that on the same held-out questions, declared scope cuts wrong returns from 20 to 3 against plain search over the same pages.

13. Conclusion

The parts were written one at a time, each for the scale in front of it, and each is right at that scale.

Underneath them is a rule short enough to say in one breath: one owner returns, two owners stop, no owner refuses, a change adds a page, a derivation adds one to a counter.

Written down as code, the rule turns out to need very little. It needs a definition of ownership that is a set comparison and not a ranking. It needs the hash to leave out the one field that has to change. It needs the new page written before the old one is retired. It needs a revision to inherit depth from what it replaces. None of that is interesting, which the swarm paper already identified as the intended outcome.

The test the formula asked for was run, found a bug, and was run again. Stripping the fields never removed the refusal; it removed the reader's ability to refuse the right things. With the bug fixed, the fields make the reader right twice as often as the same code without them, and wrong a small fraction as often as search.

They do not make it never wrong. Ask about the gas kiln from an angle no page covers and the gas kiln page comes back. A ledger with a refusal refuses what it has no words for. It does not yet refuse what it has the words for and not the answer, and nothing in a rule over words will teach it to. That is the next thing to build, and it belongs on the page, not in the reader.

Appendix A — Commands

Each is run as `python3 -m mplpb_combined <command>`.

```
ask      ROOT "question" [--profile lab|internal|external] [--no-prose] [--why] [--json]
validate ROOT [--strict]
list     ROOT [--all]
show     ROOT ID
history  ROOT ID
write    ROOT --title T --scope S [--when-to-use W] [--body B] [--prefix P]
revise   ROOT ID [--body B] [--scope S] [--ratified-by NAME]
derive   ROOT --from ID [--from ID] --title T --scope S
ratify   ROOT ID --who NAME
withdraw ROOT ID
seal     ROOT
index    ROOT
killtest ROOT PROBES.json [--out results.json]
```

`ask` exits 0 for a return, 2 for ambiguous, 3 for not in corpus. `validate` exits 1 on any error.

Appendix B — A Page

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Bisque firing schedule</title>
<meta name="mplpb:document-id" content="STUDIO-0002">
<meta name="mplpb:scope" content="Bisque firing schedule for greenware">
<meta name="mplpb:when-to-use" content="first firing; how fast to heat raw clay">
<meta name="mplpb:status" content="current">
<meta name="mplpb:hash" content="sha256:...">
<meta name="mplpb:origin" content="human">
<meta name="mplpb:origin-depth" content="0">
<meta name="mplpb:supersedes" content="STUDIO-0001@sha256:...">
</head>
<body>
<h1>Bisque firing schedule</h1>
<p>Load bone-dry ware only. Climb 100 C per hour to 600 C ...</p>
</body>
</html>
```

Appendix C — Validation Checks

Check	Level	What it finds
C1	error	A required field is missing
C2	error	Content does not match its hash
C3	error	Two pages hold one id
C4	error	Status or origin has an unknown value
C5	error	A reference names a page not in the folder
C6	error	A pinned parent no longer has the pinned hash
C7	warning	A reference carries no hash
C8	error	Declared depth disagrees with lineage
C9	warning	A superseded page still says current
C10	error	Two current pages supersede the same predecessor
C11	error or warning	A pointer has no target, or its target is not reachable
C12	error	Lineage loops
C13	warning	A file under the root is not a record
C14	error or warning	The log chain is broken, or a page is not in the log

Appendix D — Revision History

v2 supersedes **v1** the same day. **v1** is retained under docs/superseded/, its reader at killtest/v1/reader_v1.py, and its results at killtest/v1/.

Corrections that changed conclusions:

- **v1** said the reader with declared scope "returned no wrong page" and built its reading on that. On held-out questions the same reader returns four. The statement was true of 56 questions and is withdrawn as a general one.
- **v1** defined ownership as declaring every word any page declared, and matched no prose. Both are replaced (section 3). docs/FIXLOG.md entries E1 to E5 give error, cause, fix and rerun for each change.
- **v1**'s coverage dial and common-word rule are removed.

Two harness changes made during **v1** remain as they were: the summary row excludes only circular classes, and possessives are stripped before tokenising.

Source Note

Code is MIT. Documentation is CC BY 4.0. Each part under parts/ keeps its own licence file.

This paper and the mplpb_combined package were drafted with Claude (Anthropic) from the owner's formula, the uploaded papers, and the four public repositories, on 2026-10-07. Every figure in sections 8

and 9 was produced by running the included code, and the shipped results for both probe sets are checked against fresh runs by the test suite. No web search was available during drafting; nothing here draws on sources beyond the supplied material.

Building the PDF needs `report lab`. That is a documentation dependency. It is not imported by the package and is not needed to run it or its tests.

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